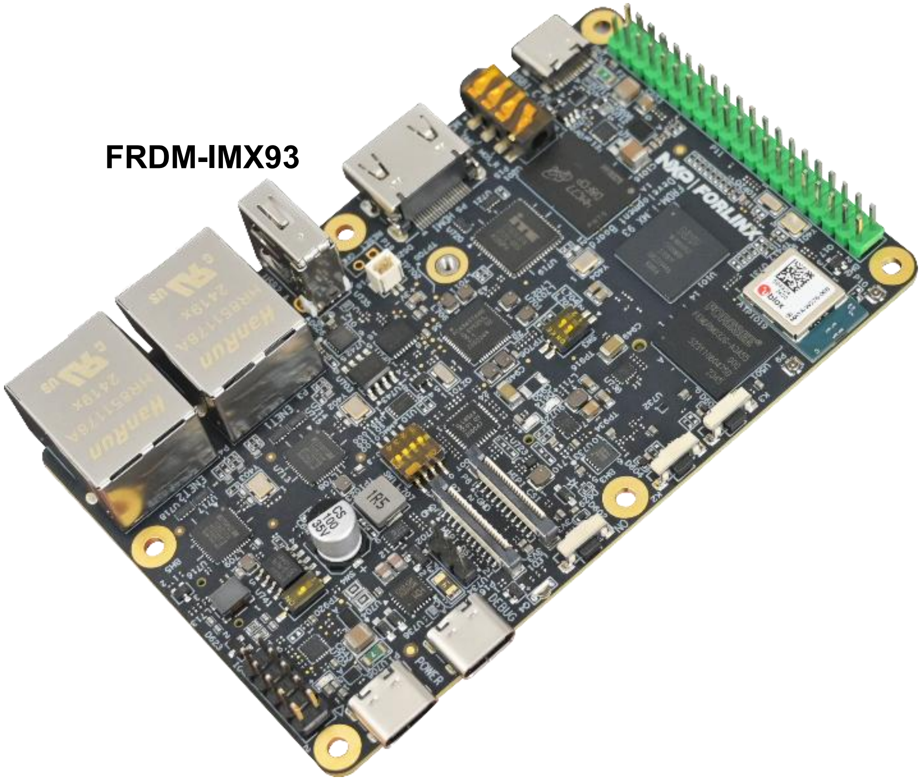
MakeNTU NXP FRDM-IMX93 Introduction

Mar.2026

AVNET
Reach Further™

Agenda

FRDM-IMX93



HW overview
- i.MX 93 introduction
- FRDM-IMX93 Features

SW environment
- Flash image
- Build Embedded Linux

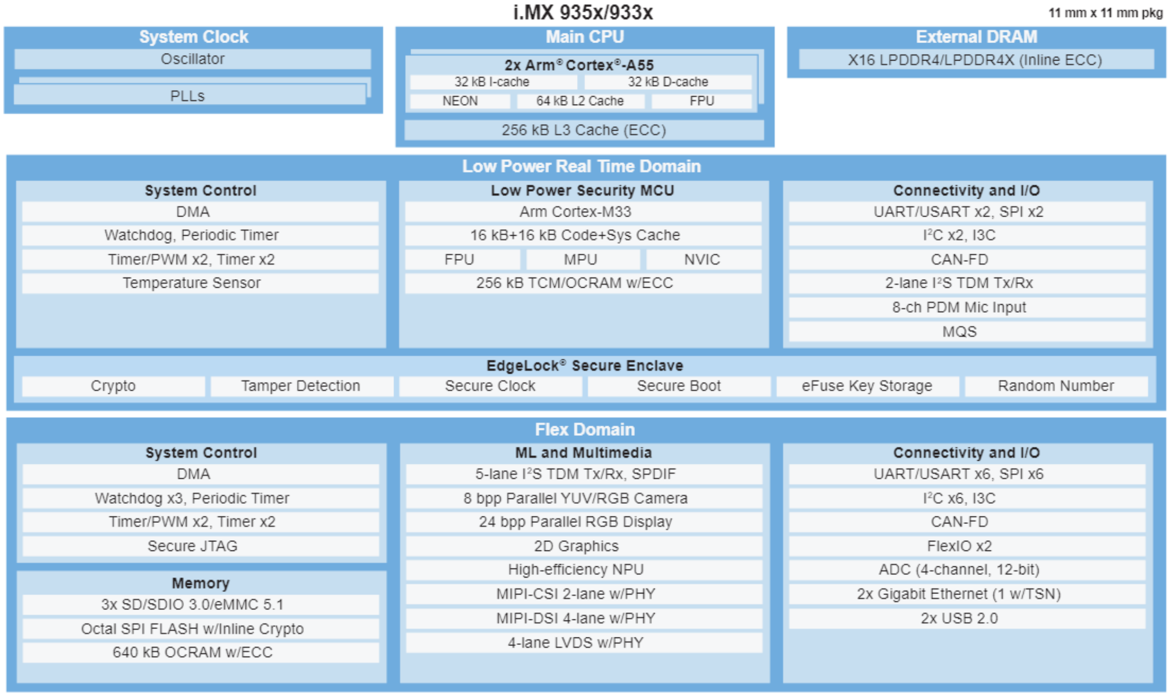
Working with accessories
- Use USB CAM
- Enable WiFi
- Basic IO control

GoPoint / NPU

HW overview

i.MX93 - Block Diagram

AVNET



Operating Systems
- Linux® OS , FreeRTOS , QNX

i.MX 93 Applications Processors

Multicore Processing
- Arm® Cortex®-A55 @ 1.7 GHz (x2)
- Arm® Cortex®-M33 @ 250Mhz
- Arm® Ethos™ U-65 microNPU
- EdgeLock® secure enclave

External Memory
- Up to 3.7GT/s x16 LPDDR4/LPDDR4X
- 3x SD 3.0/ SDIO3.0/ eMMC5.1
- Octal SPI, including support for SPI NOR and SPI NAND memories

Graphics
- Hardware Compositor for Blending/Composition, Resize, Color Space Conversion

Display Interfaces
- 1080p60 MIPI-DSI (4-lane, 1.5Gbps/lane)
- 720p60 LVDS (4-lane)
- 24-bit parallel RGB

Camera Interfaces
- 1080p60 MIPI-CSI (2-lane, 1.5Gbps/lane)
- 8-bit parallel YUV/RGB

Audio Interfaces
- 7x I2S TDM (32-bit @ 768KHz), SPDIF Tx/Rx
- 8 channel PDM microphone input
- MQS: Medium Quality Sound output

Get to know **FRDM i.MX 93** Board

2x **Ethernet** ports

MIPI-CSI, MIPI-DSI and HDMI connectors

2x20 header for **expansion boards** and **adapters**

Debug, PD and USB type-C + **USB** Type-A

IW612 Wireless Module - **u-blox: MAYA-W276**

- **TRI-RADIO** Wi-Fi 6 + Bluetooth + 802.15.4
- NXP EdgeLock® Security
- Matter over Wi-Fi & Thread
- Thread Border Router
- Thread or Zigbee
- BDR, EDR and Bluetooth LE

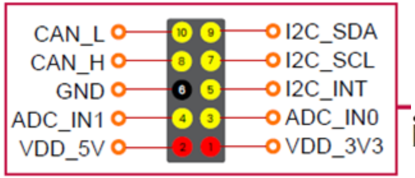
Specifications

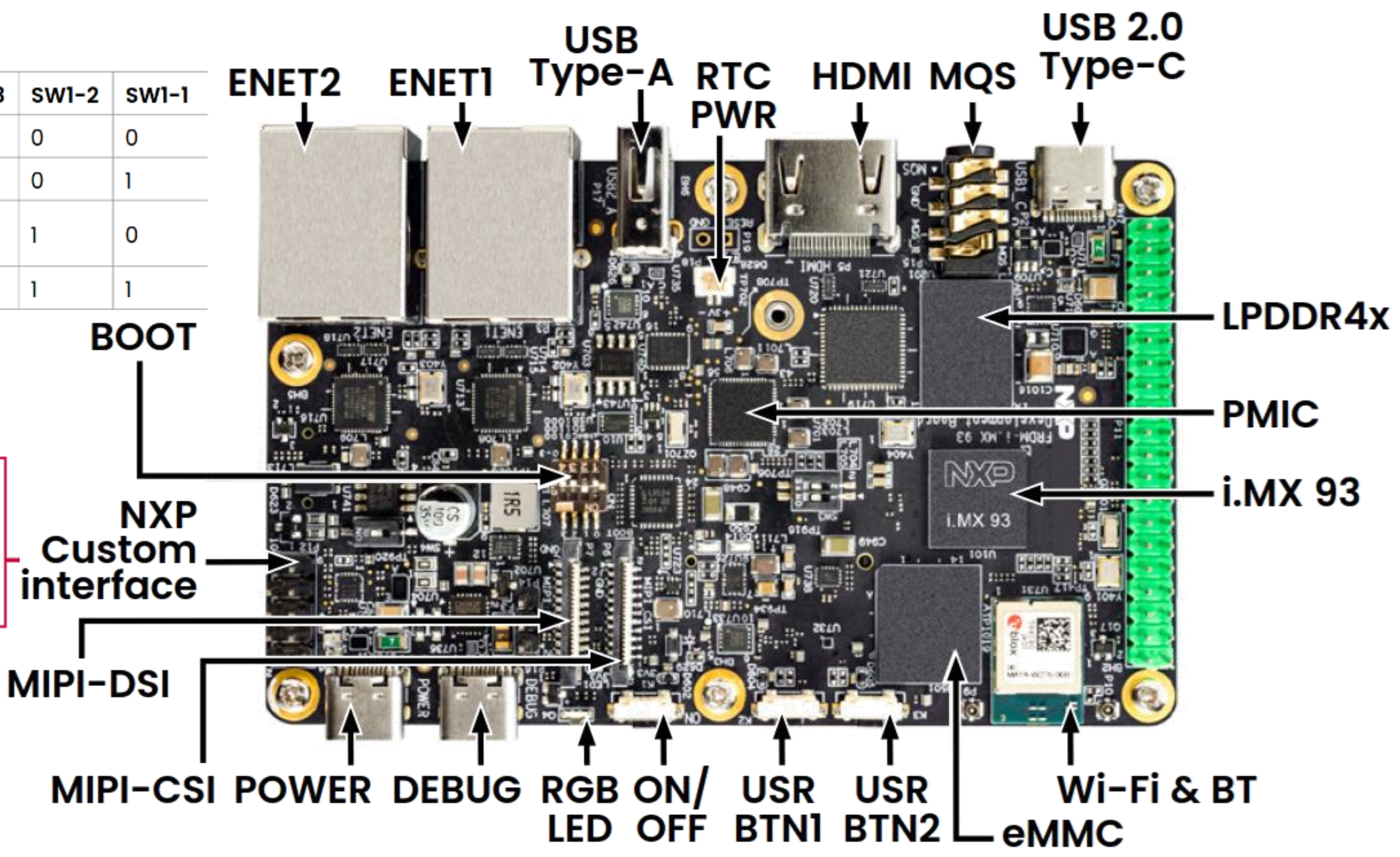
- 2x Arm Cortex®-A55 + Cortex®-M33
- Wi-Fi 6 + BT + 802.15.4 Module on-board, IW612
- 2x GB Ethernet (1xETER, 1xTSN)
- RPI MIPI-CSI/DSI, HDMI
- M.2 Connector
- LPDDR4X 16-bit 2GB
- eMMC 5.1, 32GB
- MicroSD 3.0 card slot
- 3x USB 2.0 Type-C connector (one for Debug, one PD only) + 1x USB 2.0 Type-A
- RTC, Buttons and LED

NXP Devices On-Board	
PMIC	PCA9451A
USB PD TCPC PHY IC	PTN5110
High-Voltage USB PD Power Switch	NX20P5090UK
IIC Extends GPIO	PCAL6524/PCAL6408A
CAN Transceiver	TJA1051
USB Sink & Source combo power switch	NX20P3483UK
USB Type-C CC and SBU Protection IC	NX20P0407
Real-time clock/calendar	PCF2131
Wi-Fi, BT, 802.15.4 Tri-Radio	IW612 (in u-blox Module)

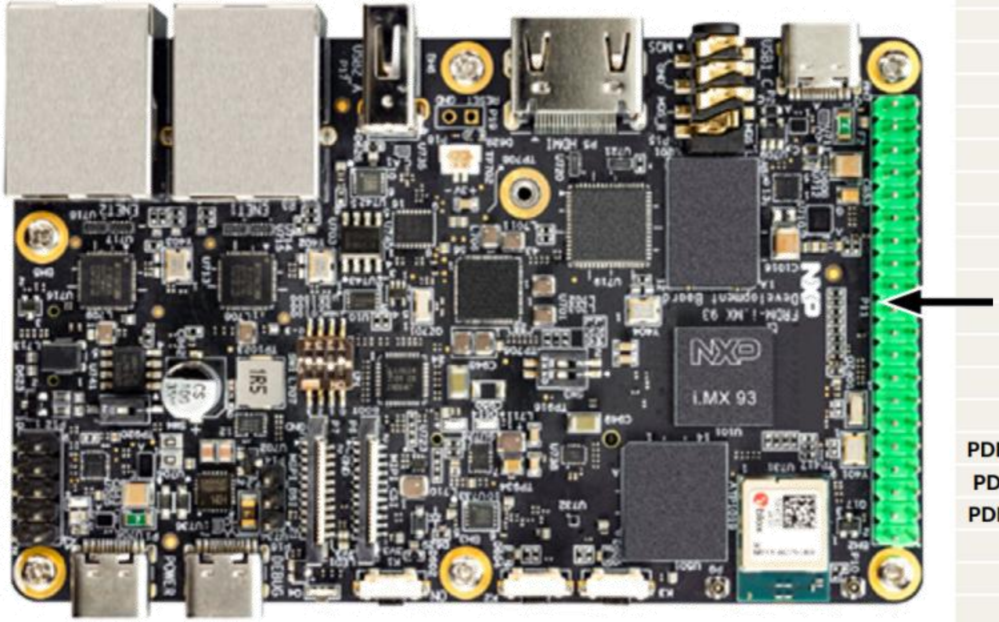
FRDM i.MX93 Board - Front

BOOT mode	SW1-4	SW1-3	SW1-2	SW1-1
From internal fuses	0	0	0	0
Serial downloader	0	0	0	1
USDHC1 8-bit eMMC 5.1	0	0	1	0
USDHC2 4-bit SD3.0	0	0	1	1



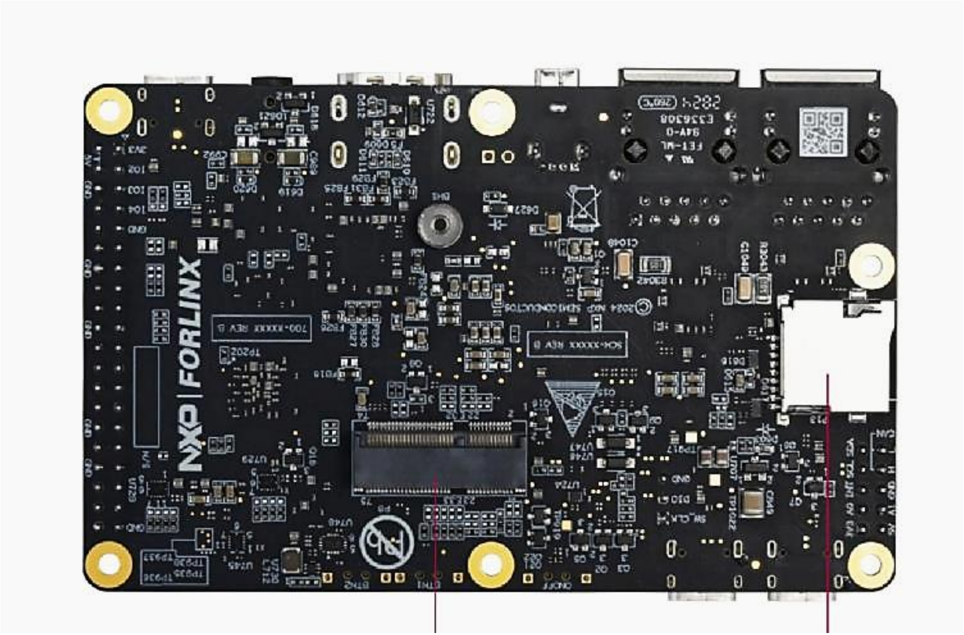


FRDM i.MX93 Board - Front



	3V3	1	2	5V	
I2C4.SDA	GPIO_2	3	4	5V	
I2C4.SCL	GPIO_3	5	6	GND	
PDM.CLK	GPIO_4	7	8	GPIO_14	UART3.TXD
	GND	9	10	GPIO_15	UART3.RXD
	GPIO_17	11	12	GPIO_18	SAI3.RXBCLK
	GPIO_27	13	14	GND	
	GPIO_22	15	16	GPIO_23	
	3V3	17	18	GPIO_24	
SPI3.SOUT	GPIO_10	19	20	GND	
SPI3.SIN	GPIO_9	21	22	GPIO_25	
SPI3.SCK	GPIO_11	23	24	GPIO_8	SPI3.CS0
	GND	25	26	GPIO_7	SPI3.CS1
	GPIO_0	27	28	GPIO_1	
PDM.BITSTREAM_0	GPIO_5	29	30	GND	
PDM.BITSTREAM_1	GPIO_6	31	32	GPIO_12	PDM.BITSTREAM_2
PDM.BITSTREAM_3	GPIO_13	33	34	GND	
SAI3_RXSYNC	GPIO_19	35	36	GPIO_16	SAI3.TXBCLK
SPI3.TXSYNC	GPIO_26	37	38	GPIO_20	SAI3.RXDATA0
	GND	39	40	GPIO_21	SAI3.TXDATA0

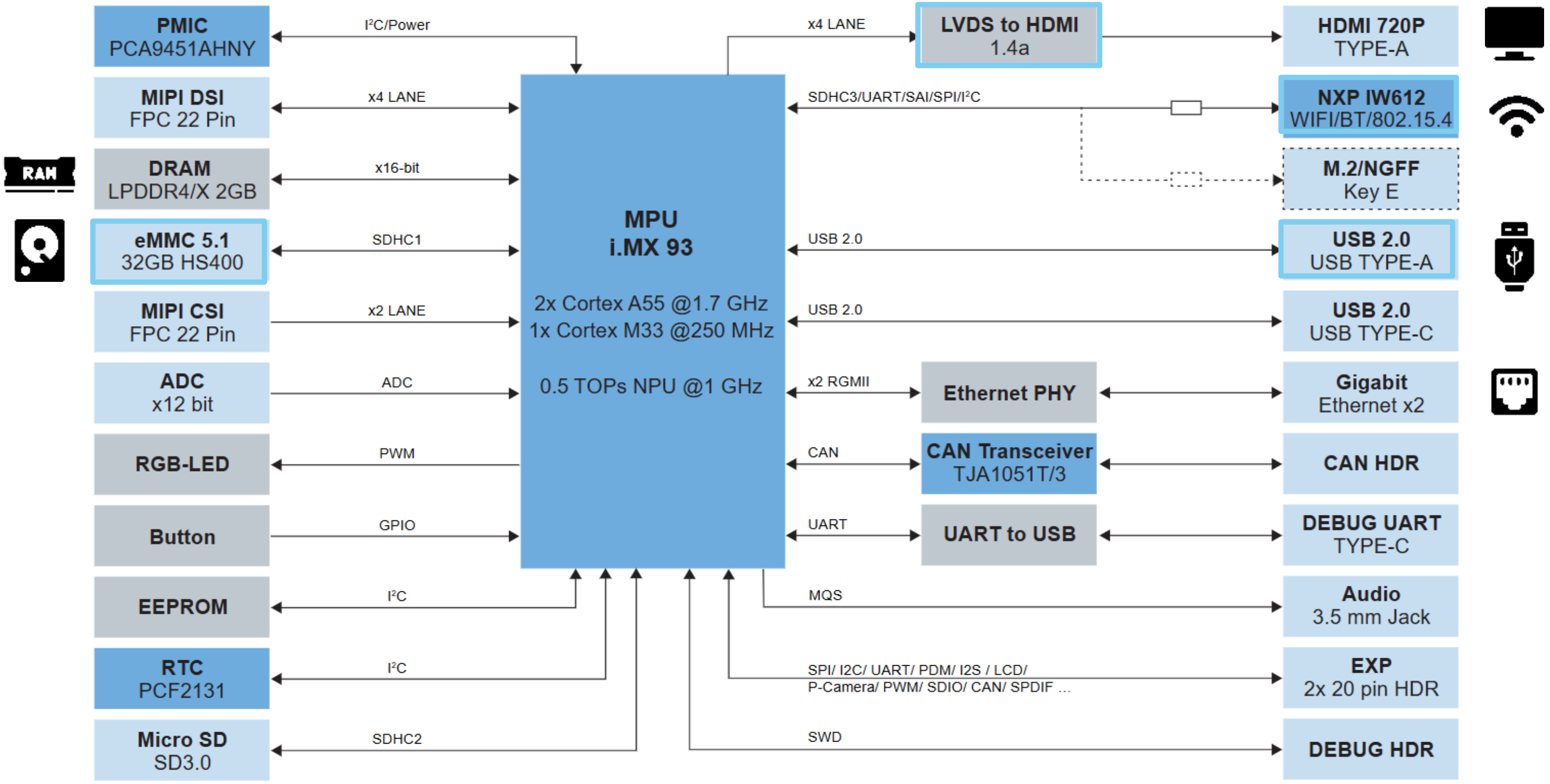
FRDM i.MX93 Board - Back



M.2 slot

MicroSD slot

FRDM i.MX93 Board - Block Diagram



NXP components | Headers and connectors | On-board components | Rework required

圖式出處：

SW environment

FRDM-IMX93 SW Release Package

Yocto Release

- https://github.com/nxp-imx/imx-manifest

Debian Release

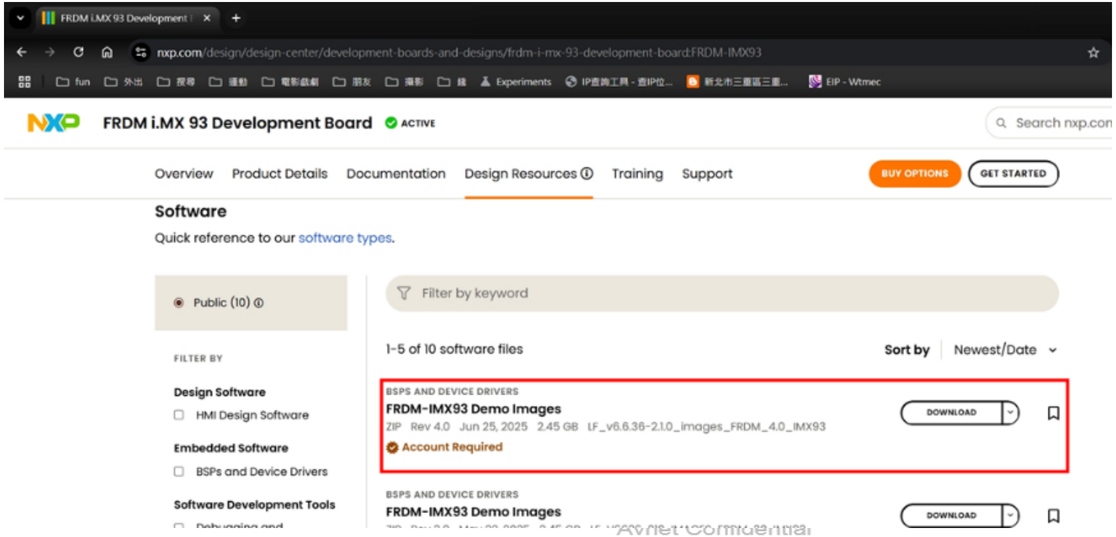
- https://github.com/nxp/flexbuild

Getting Started with FRDM i.MX93

Step 1

Download Pre-Build Image (Yocto Linux)

https://www.nxp.com/design/design-center/development-boards-and-designs/frdm-i-mx-93-development-board:FRDM-IMX93

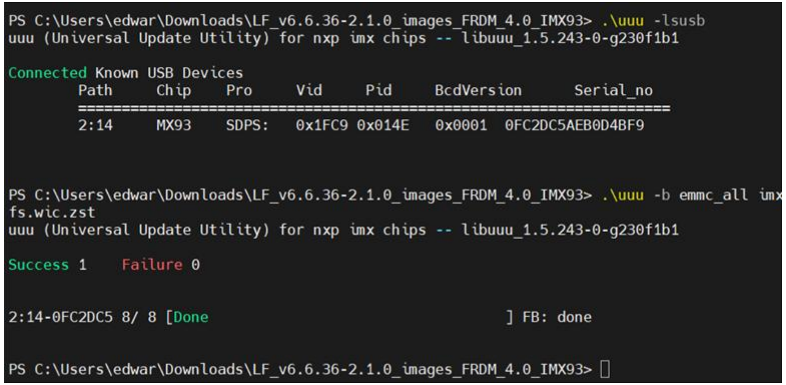


Getting Started with FRDM i.MX93

AVNET

Step 2

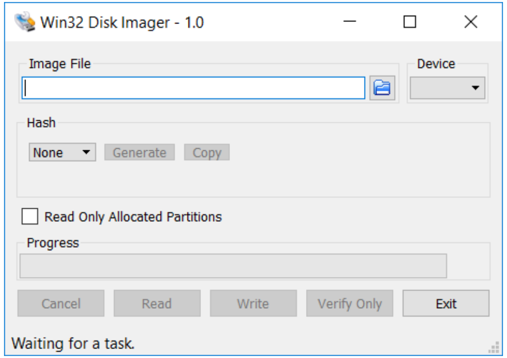
1. Using uuu to burn image and rootfs to SDCard/EMMC:
   - Download the latest stable files from [UUU GitHub page](#).
   - //Linux
   - $ chmod a+x uuu
   - $ sudo ./uuu -b sd_all imx-image-full-imx93frdm.rootfs.wic.zst
   - Or $ sudo ./uuu -b emmc_all imx-image-full-imx93frdm.rootfs.wic.zst
   - //Windows
   - > ./uuu.exe -b sd_all imx-image-full-imx93frdm.rootfs.wic.zst
   - Or > ./uuu.exe -b emmc_all imx-image-full-imx93frdm.rootfs.wic.zst



OR

2. Flashing an SD card image by linux
   - $ zstdcat imx-image-full-imx93frdm.rootfs.wic.zst | sudo dd of=/dev/sdx bs=1M && sync

Flashing an SD card image by Windows Win32Diskmanager
https://win32diskimager.org/

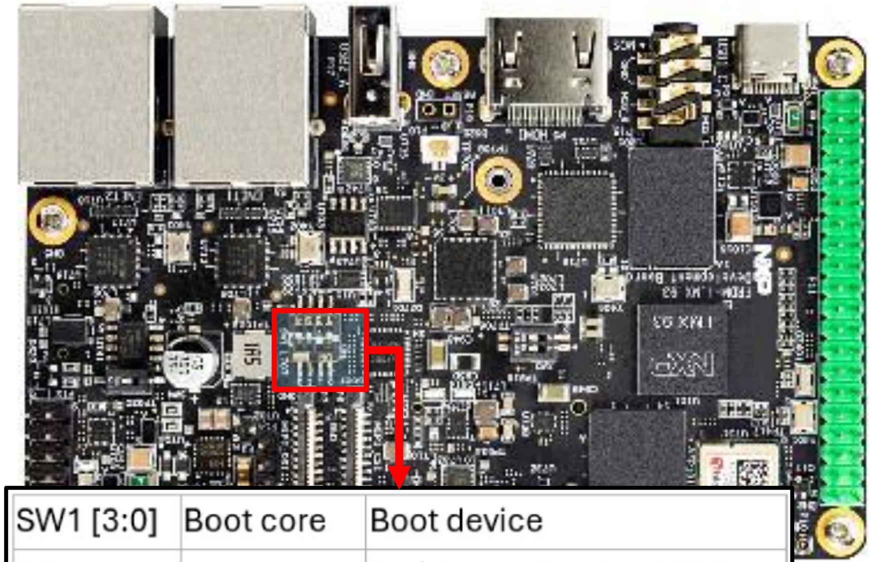


Getting Started with FRDM i.MX93

Step 2 Backup

```
PS C:\Users\edwar\Downloads\LF_v6.6.36-2.1.0_images_FRDM_4.0_IMX93> .\uuu -lsusb
uuu (Universal Update Utility) for nxp imx chips -- libuuu_1.5.243-0-g230f1b1

Connected Known USB Devices
        Path     Chip     Pro      Vid      Pid      BcdVersion      Serial_no
        ====================================================================
        2:14     MX93     SDPS:    0x1FC9 0x014E    0x0001   0FC2DC5AEB0D4BF9


PS C:\Users\edwar\Downloads\LF_v6.6.36-2.1.0_images_FRDM_4.0_IMX93> .\uuu -b emmc_all imx-boot-imx93frdm-sd.bin-flash_singleboot imx-image-full-imx93frdm.root
fs.wic.zst
uuu (Universal Update Utility) for nxp imx chips -- libuuu_1.5.243-0-g230f1b1

Success 1    Failure 0


2:14-0FC2DC5 8/ 8 [Done                                  ] FB: done


PS C:\Users\edwar\Downloads\LF_v6.6.36-2.1.0_images_FRDM_4.0_IMX93>
```

Getting Started with FRDM i.MX93



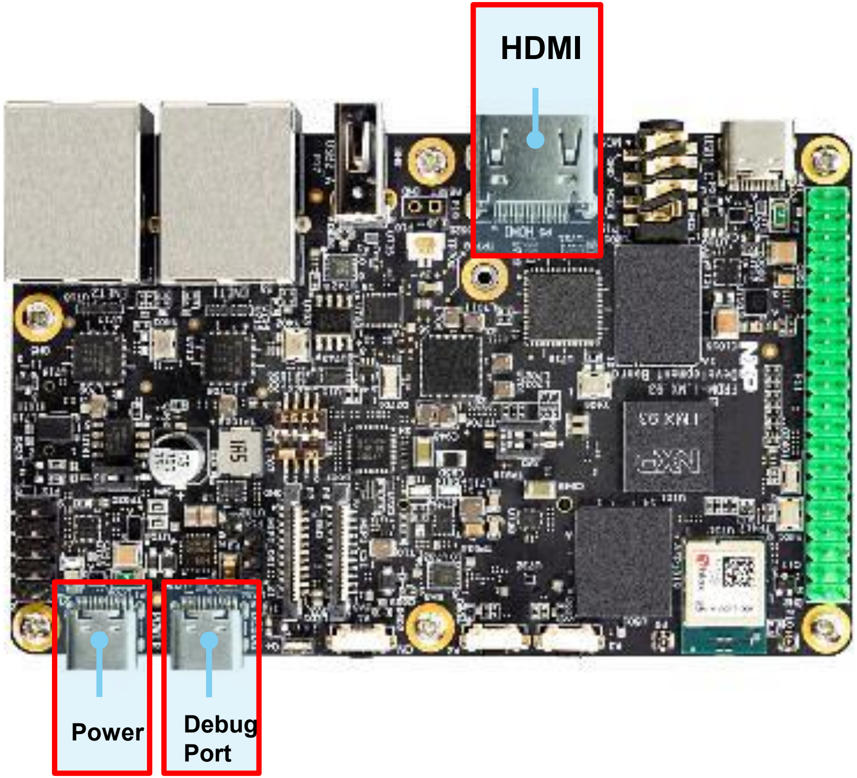
SW1 [3:0]	Boot core	Boot device
0001		Serial downloader (USB)
0010	Cortex-A55	uSDHC1 8-bit eMMC 5.1
0011		uSDHC2 4-bit SD3.0

Step 3

Boot switch setup to 0011(SD boot)

or 0010(EMMC boot) recommend SD boot.

Getting Started with FRDM i.MX93



Step 4

Connect the power supply, HDMI cable, debug port and SD Card to the FRDM board.

After powering on, wait for a few seconds. If you see the penguin icon in the top left corner, the boot process is successful.

FRDM-IMX93 Yocto Release - BSP

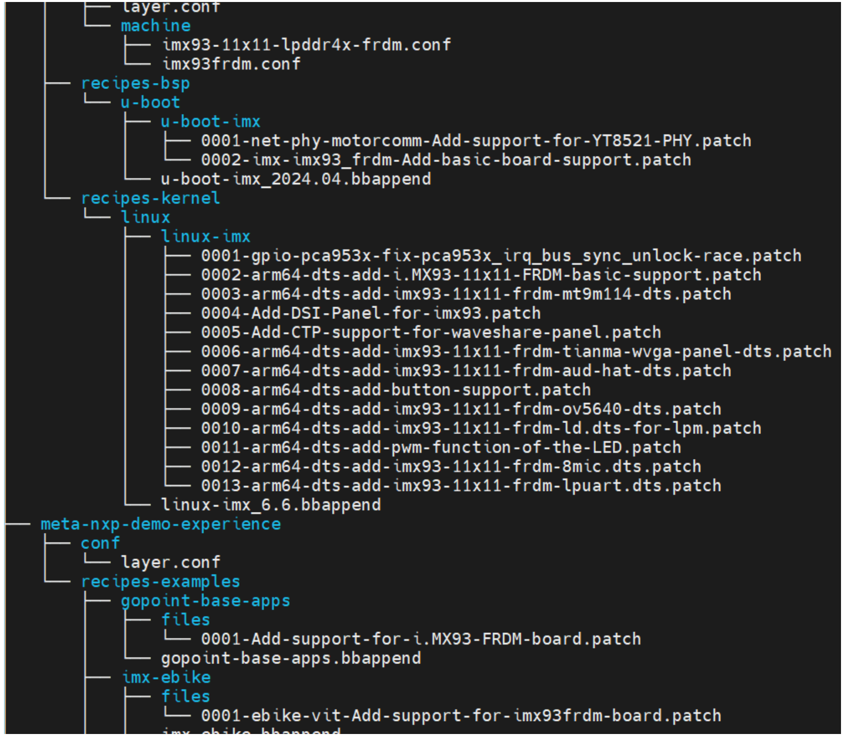
Based on i.MX SW 2024 Q3 release

- Linux kernel: 6.6.36
- u-boot: 2024.04

Source: https://github.com/nxp-imx-support/meta-imx-frdm

FRDM-IMX93 BSP changes:

- U-boot:
  - Add basic support for FRDM-IMX93;
- Kernel:
  - Add basic support for FRDM-IMX93;
  - Add support for kinds of accessories;
- GoPoint:
  - Add FRDM-IMX93 support;
- FRDM-IMX93 Yocto layer:
  - Add Yocto layer for FRDM-IMX93;
  - Integrate u-boot/kernel/GoPoint patches;



```
│   │   ├── layer.conf
│   │   └── machine
│   │       ├── imx93-11x11-lpddr4x-frdm.conf
│   │       └── imx93frdm.conf
│   ├── recipes-bsp
│   │   └── u-boot
│   │       ├── u-boot-imx
│   │       │   ├── 0001-net-phy-motorcomm-Add-support-for-YT8521-PHY.patch
│   │       │   └── 0002-imx-imx93_frdm-Add-basic-board-support.patch
│   │       └── u-boot-imx_2024.04.bbappend
│   └── recipes-kernel
│       └── linux
│           ├── linux-imx
│           │   ├── 0001-gpio-pca953x-fix-pca953x_irq_bus_sync_unlock-race.patch
│           │   ├── 0002-arm64-dts-add-i.MX93-11x11-FRDM-basic-support.patch
│           │   ├── 0003-arm64-dts-add-imx93-11x11-frdm-mt9m114-dts.patch
│           │   ├── 0004-Add-DSI-Panel-for-imx93.patch
│           │   ├── 0005-Add-CTP-support-for-waveshare-panel.patch
│           │   ├── 0006-arm64-dts-add-imx93-11x11-frdm-tianma-wvga-panel-dts.patch
│           │   ├── 0007-arm64-dts-add-imx93-11x11-frdm-aud-hat-dts.patch
│           │   ├── 0008-arm64-dts-add-button-support.patch
│           │   ├── 0009-arm64-dts-add-imx93-11x11-frdm-ov5640-dts.patch
│           │   ├── 0010-arm64-dts-add-imx93-11x11-frdm-ld.dts-for-lpm.patch
│           │   ├── 0011-arm64-dts-add-pwm-function-of-the-LED.patch
│           │   ├── 0012-arm64-dts-add-imx93-11x11-frdm-8mic.dts.patch
│           │   └── 0013-arm64-dts-add-imx93-11x11-frdm-lpuart.dts.patch
│           └── linux-imx_6.6.bbappend
├── meta-nxp-demo-experience
│   ├── conf
│   │   └── layer.conf
│   └── recipes-examples
│       ├── gopoint-base-apps
│       │   ├── files
│       │   │   └── 0001-Add-support-for-i.MX93-FRDM-board.patch
│       │   └── gopoint-base-apps.bbappend
│       ├── imx-ebike
│       │   ├── files
│       │   │   └── 0001-ebike-vit-Add-support-for-imx93frdm-board.patch
```

FRDM-IMX93 Yocto Release Usage

Download i.MX Yocto Release:

- $ mkdir imx-yocto-bsp
- $ cd imx-yocto-bsp
- $ repo init -u https://github.com/nxp-imx/imx-manifest -b imx-linux-scarthgap -m imx-6.6.36-2.1.0.xml
- $ repo sync

Integrate i.MX FRDM layer into i.MX Yocto Project:

- $: cd ./sources
- $: git clone https://github.com/nxp-imx-support/meta-imx-frdm.git
- $: cd meta-imx-frdm
- $: git checkout imx-frdm-4.0

Yocto Project Setup:

- $ MACHINE=imx93frdm DISTRO=fsl-imx-xwayland source sources/meta-imx-frdm/tools/imx-frdm-setup.sh -b frdm-imx93

Build images:

- $ bitbake imx-image-full

Flashing an SD card image:

- $ zstdcat imx-image-full-imx93frdm.rootfs.wic.zst | sudo dd of=/dev/sdb bs=1M && sync

Using uuu to burn image and rootfs to SD:

- $ uuu -b sd_all imx-boot-imx93frdm-sd.bin-flash_singleboot <rootfs.wic.zst>

FRDM-MX93 Debian Release

Debian is a free Operating System (OS), also known as Debian GNU/Linux.

i.MX Debian Linux SDK distribution is a combination of NXP-provided kernel and boot loaders with a Debian distro user-space image.

- Debian 13
- NXP packages are based i.MX SW Release 2025 Q4

i.MX Debian Linux SDK distribution uses Flexbuild to build system.

- Debian-based RootFS;
  - Debian Base (basic packages)
  - Debian Server (more packages without GUI Desktop)
  - Debian Desktop (with GNOME GUI Desktop)
- Linux kernel;
- BSP components;
- various applications (graphics, multimedia, networking, connectivity, security, and AI/ML);

Source: https://github.com/NXP/flexbuild

Introduction: https://nxp.com/nxpdebian

Quick Start with Debian

Flexbuild compiles and assembles the distro images as three parts:

- BSP firmware image
- Boot image
- RootFS image

Creating an SD card on the Linux host

- Download flex-installer
  - $ wget http://www.nxp.com/lgfiles/sdk/lsdk2512/flex-installer
  - $ chmod +x flex-installer; sudo mv flex-installer /usr/bin
- Plug the SD card into the Linux host and install the images as below:
  - $ flex-installer -i pf -d /dev/**sdb** (format SD card)
  - $ flex-installer -i auto -d /dev/**mmcblk1** -m **imx93frdm** (automatically download and install images)
- Plug the SD card into the i.MX board and install the extra packages as follows:
  - $ dhclient -i end0 (setup Ethernet network interface by DHCP or setting it manually)
  - $ date -s "28 Mar 2026 09:00:00" (setting correct system time is required)
  - $ debian-post-install-pkg **desktop** (install extra packages for GNOME GUI Desktop version)
  - or
  - $ debian-post-install-pkg **server** (install extra packages for Server version without GUI Desktop)
  - # After finishing the installation, run the reboot command to boot up the Debian Desktop/Server system.

Building Debain with FRDM i.MX93

(1) Download flex-installer :

```
$ wget http://www.nxp.com/lgfiles/sdk/lsdk2512/flex-installer
$ chmod +x flex-installer; sudo mv flex-installer /usr/bin
```

(2) Download FlexBuild Release:

```
$ git clone https://github.com/nxp/flexbuild
```

(3) Install Docker

```
$ sudo apt install docker.io
```

(4) Building Debian images in Flexbuild:

```
$  cd flexbuild
$ . setup.env  (in host environment)
$ bld docker   (create or attach to docker)
$ . setup.env  (in docker environment)
$ bld host-dep (install host dependent packages)
$ bld -m imx93frdm
```

(4) Burn Image to SD Card:

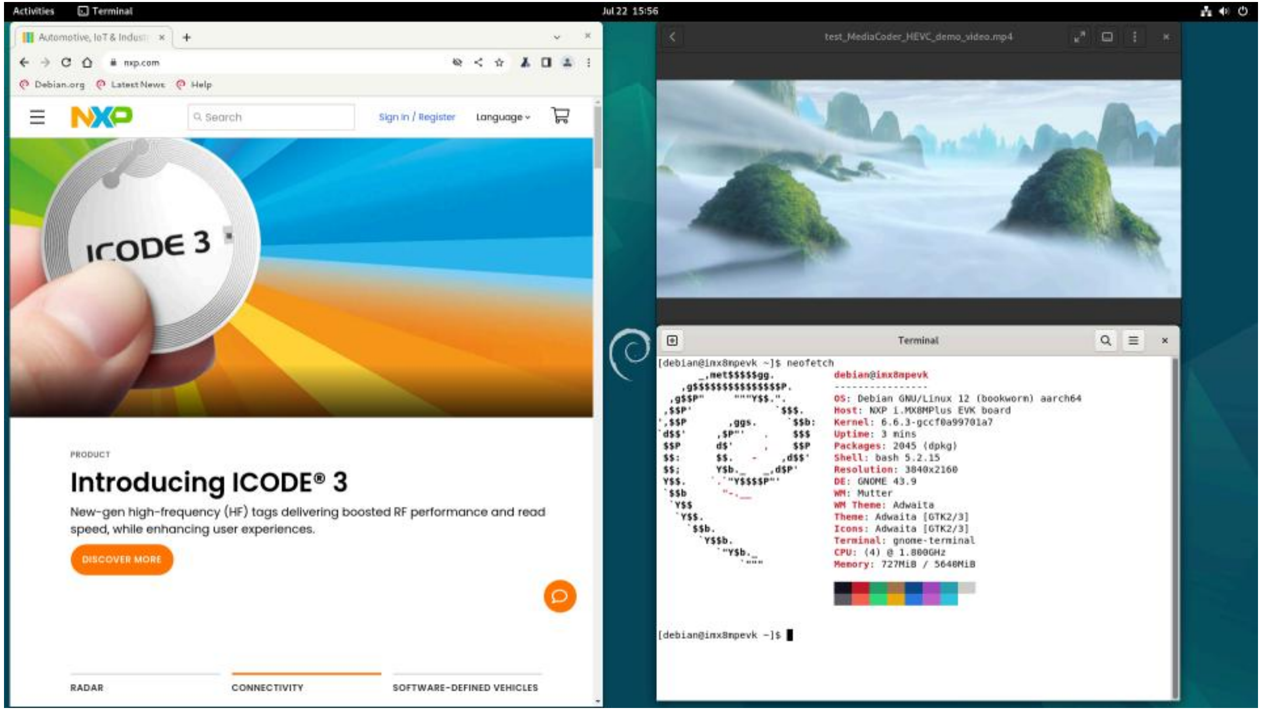
```
$ cd build_lsdk2512/images
$ flex-installer -i pf -d /dev/mmcblkX
$ flex-installer -d /dev/mmcblk1 -m imx93frdm \
        -f firmware_imx93frdm_sdboot.img \
        -b boot_IMX_arm64_lts_6.12.20 \
        -r rootfs_lsdk2512_debian_imx93frdm.tar.zst
```

Building Debian Images with Flexbuild usage

Flexbuild usage:

- $ bld -m imx93frdm (build all images for imx93frdm)
- $ bld uboot -m imx93frdm (compile u-boot image for imx93frdm)
- $ bld linux (compile linux kernel for all arm64 i.MX machines)
- $ bld bsp -m imx93frdm (generate BSP firmware)
- $ bld boot (generate boot partition tarball including kernel, dtb, modules, distro bootscript for iMX machines)
- $ bld multimedia (build multimedia components for i.MX platforms)
- $ bld rfs -r debian:base (generate Debian base rootfs with base packages)
- $ bld apps -r debian:server (compile apps against runtime dependencies of Debian server RootFS)
- $ bld merge-apps -r debian:server (merge iMX-specific apps into target Debian server RootFS)
- $ bld packrfs (pack and compress target rootfs)

Debian desktop



Related Documentation

AVNET

FRDM-IMX93 Documents:

FRDM-IMX93 Quick Start Guide
FRDM-IMX93 Board User Manual
FRDM-IMX93 Software User Guide

More information about i.MX productions can be found at(http://www.nxp.com/imxlinux)

i.MX Yocto Project User’s Guide
i.MX Linux User’s Guide
i.MX Linux Reference Manual
i.MX Porting Guide

Debian documents at http://www.nxp.com/nxpdebian

i.MX Debian Linux SDK User Guide

NPU introduction

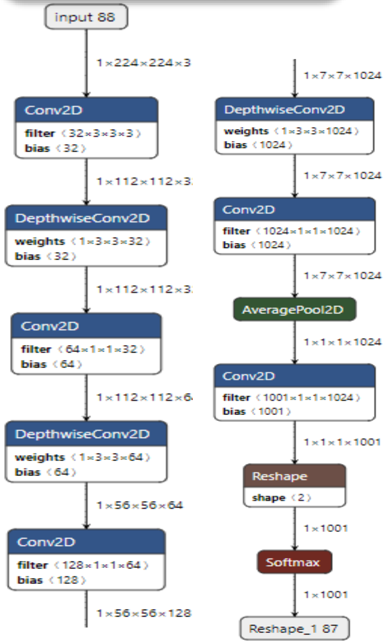
NPU with FRDM i.MX93

- **Using arm Ethos-U65 of NPU**
  (https://developer.arm.com/Processors/Ethos-U65)
- **Higher-Efficiency Performance : 0.5 TOP/s @ 1 GHz**
- **Support 8-bit integer weight operation.**
- **Support 8-bit and 16-bit integer activations operators.**
- **Support ML Framework at MPU Chip, including TensorFlow Lite.**
  (NXP i.MX8M PLUS / i.MX93 Series)
- **Non-need Warm-Up Process**

NPU with FRDM i.MX93 – Model Converter

TensorFlow Lite

Quantized Model (*.tfilte) → Vela Compile Tool → Vela Model (*-vela.tfilte)

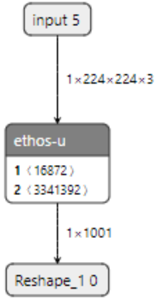


General Method :
$ vela model.tflite

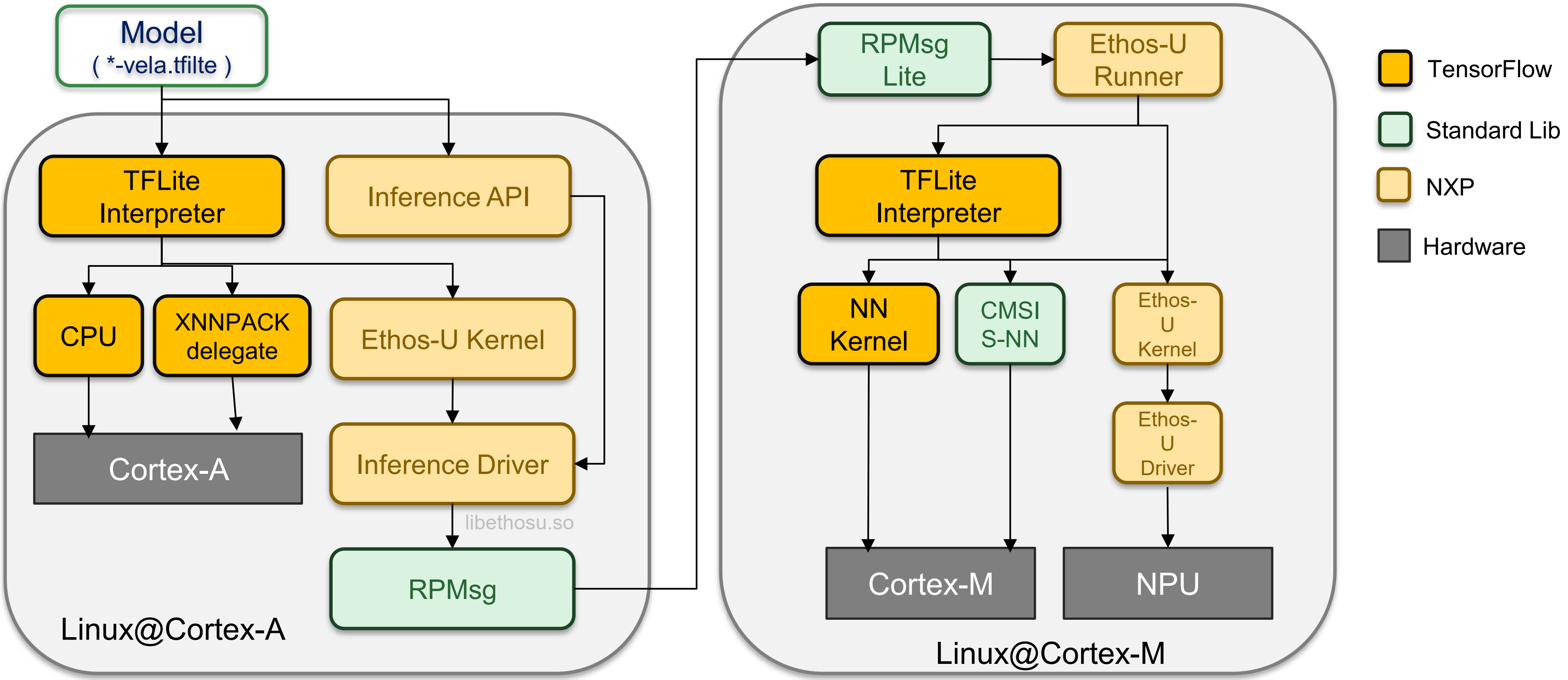
Highly-Performance Method :
$ vela --optimise Performance model.tflite

Minimum-Size Method :
$ vela --optimise Size model.tflite

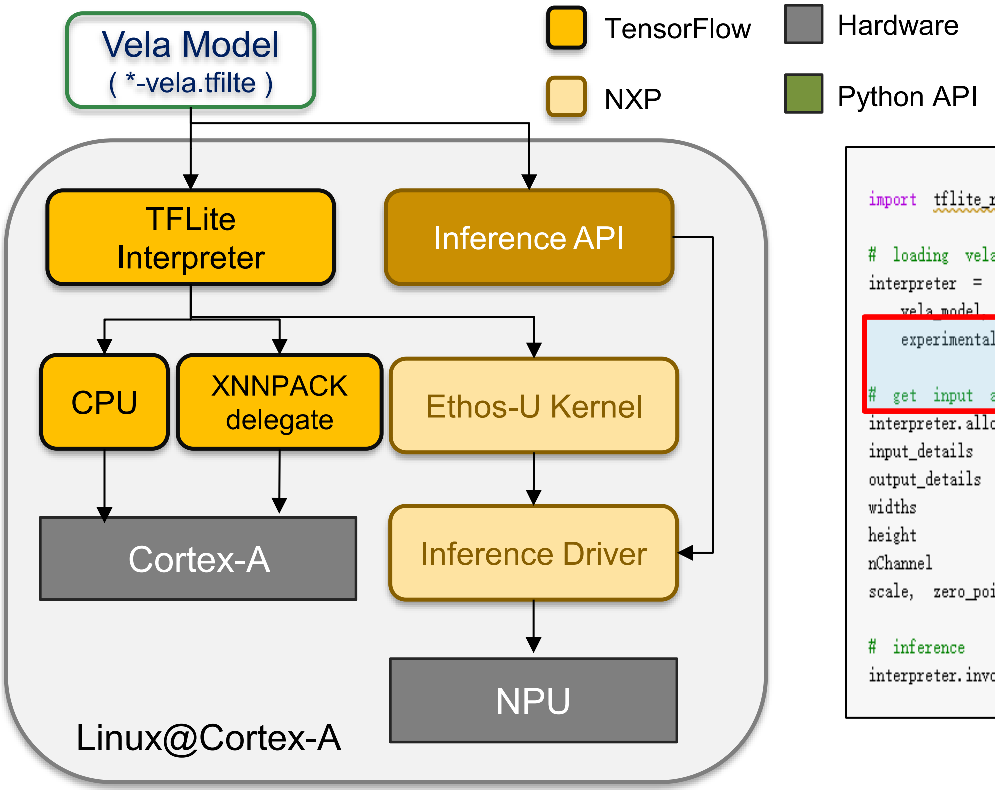
Help :
$ vela --help



NPU with FRDM i.MX93



NPU with FRDM i.MX93



How to use NPU by Python

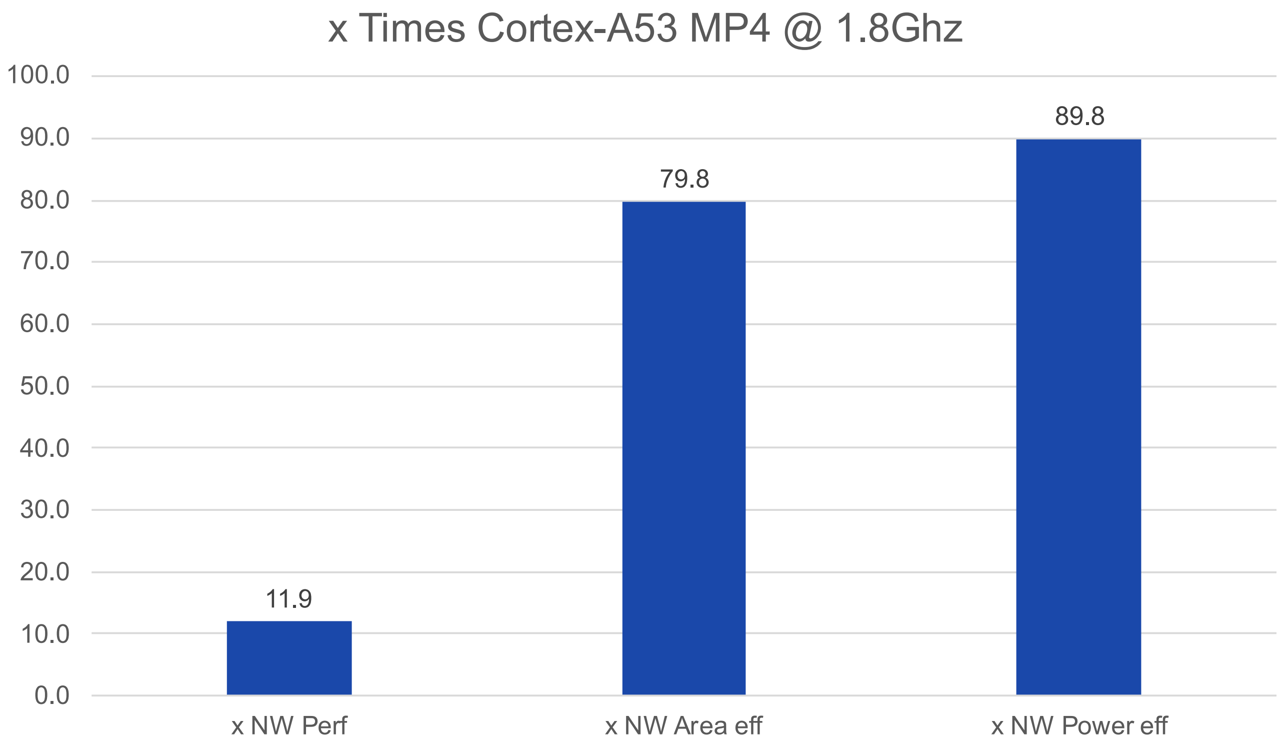
```python
import tflite_runtime.interpreter as tflite

# loading vela model
interpreter = tflite.Interpreter(
    vela_model,
    experimental_delegates=[tflite.load_delegate("/usr/lib/libethosu_delegate.so")])

# get input and output information
interpreter.allocate_tensors()
input_details  = interpreter.get_input_details()
output_details  = interpreter.get_output_details()
widths          = input_details[0]['shape'][2]
height          = input_details[0]['shape'][1]
nChannel        = input_details[0]['shape'][3]
scale, zero_point = input_details[0]['quantization']

# inference
interpreter.invoke()
```

See Here

reference : https://www.nxp.com/docs/en/user-guide/IMX-MACHINE-LEARNING-UG.pdf

NPU with FRDM i.MX93 - Performance

i.MX93 (Ethos-U65) vs Cortex-A53



With Ethos-U65, we get

11.9x times performance

89.8x times power efficiency

Over Cortex-A53

NPU with FRDM i.MX93 - Performance

AVNET

i.MX93 (Ethos-U65) vs i.MX 8M Plus

FRDM i.MX93 Target Application

Industrial Automation	Smart Home	Smart City	Automotive
• Industrial Machine vision • Industrial scanning/printing	• Smart doorbell • Smart lock • Smart home hub/ Hue Bridge	• Smart lighting • Traffic control	• Driver Monitoring System (DMS) • Object Monitoring System (OMS)




Industrial Automation and Building Control


Industrial Automation	Building Control & Energy
• Industrial HMI • Industrial gateway • I/O control • Combines with i.MX RT1180 crossover MCU for larger gateways and IO controller	• Access control • Energy meter • Energy grid equipment • EV charging station • Environmental control • Sensor hub


Debug Terminal

Debug Terminal in Windows

Serial Communication Console Setup

The WCH USB-serial chip on FRDM-IMX93 enumerates 2 serial ports. Assume that the ports are **COM11**,COM12. The first Port(**COM11**) is for the serial console communication from Arm A55 core. The serial-to-USB drivers are available at **CH342F Windows Driver** .

https://www.wch-ic.com/downloads/CH343SER_EXE.html

Note: To determine the port number of the i.MX board virtual COM port, open the Windows device manager and find USB serial Port in Ports (COM and LPT).

Recommend terminal-emulation Windows application: **Tera Term, PuTTY**

Serial port settings to 115,200 baud rate, 8 data bits, no parity and 1 stop bit.

Debug Terminal in Linux

Serial Communication Console Setup

The serial-to-USB drivers are available at **CH342F Linux Drivers** .

https://github.com/WCHSoftGroup/ch343ser_linux

On the command prompt of the Linux host machine, run the following command to determine the port number:

ls /dev/ttyCH343USB*

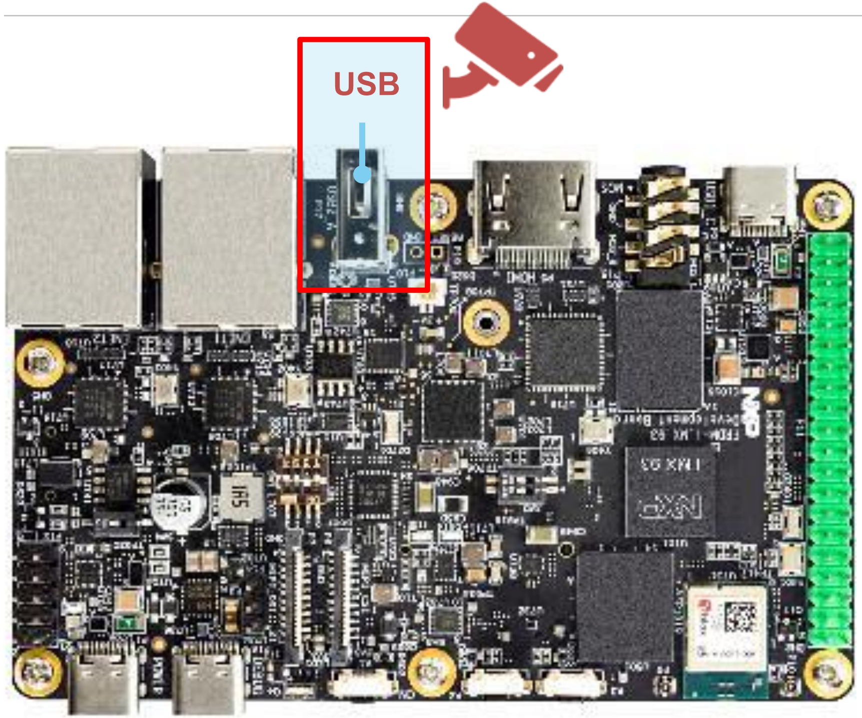
The first number is for Arm A55 core.

Recommend terminal-emulation Windows application: **Minicom**.

Serial port settings to 115,200 baud rate, 8 data bits, no parity and 1 stop bit.

Working with accessories

Camera Streaming with FRDM i.MX93



Using GStreamer to open USB Camera.

1. Connect USB Camera to FRDM board.
2. Confirm Camera Device

```
root@imx93frdm:~# v4l2-ctl --list-devices
C270 HD WEBCAM (usb-ci_hdrc.1-1):
        /dev/video0
        /dev/video1
        /dev/media0
```

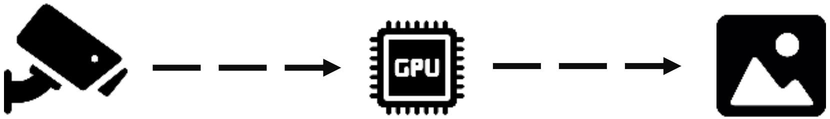
3. Running Camera

```
root@imx93frdm:~# gst-launch-1.0 v4l2src
device=/dev/video0 ! fpsdisplaysink

Setting pipeline to PAUSED ...
Pipeline is live and does not need PREROLL ...
Pipeline is PREROLLED ...
Setting pipeline to PLAYING ...
New clock: GstSystemClock
Redistribute latency...
Total showed frames (3401), playing for
(0:01:57.471764157), fps (28.952).
```


圖式出處：https://www.flaticon.com/legal

Camera Streaming with 2D GPU

AVNET



720 p

2D GPU accelerator (pxp)

1080 p

Utilize GStreamer in combination with the GPU to accelerate video processing.

root@imx93frdm:~# **gst-launch-1.0 v4l2src device=/dev/video0 ! image/jpeg, width=1280, height=720, framerate=30/1 ! jpegdec ! queue ! imxvideoconvert_pxp ! video/x-raw, width=1920, height=1080 ! waylandsink**

Setting pipeline to PAUSED ...
Pipeline is live and does not need PREROLL ...
Pipeline is PREROLLED ...
Setting pipeline to PLAYING ...
New clock: GstSystemClock
Redistribute latency...
Total showed frames (138), playing for (0:00:11.492181461), fps (12.008).

WiFi

Bring up Wi-Fi interfaces

1. Power ON the board in the internal boot mode and wait until the board is completely booted.

2. To login into the Linux running on the FRDM i.MX 93, use “**root**” as the default login username for the i.MX Linux OS. There is no password.

```
NXP i.MX Release Distro 6.6-scarthgap imx93frdm-iwxxx-matter
ttyLP0

imx93frdm-iwxxx-matterlogin: root
root@imx93frdm-iwxxx-matter:~#
```

WiFi

Bring up Wi-Fi interfaces

3. Verify the module parameter configuration file with below given configuration using the nano editor tool.

```
nano /lib/firmware/nxp/wifi_mod_para.conf
```

```
SDIW612 = {
        cfg80211_wext=0xf
        max_vir_bss=1
        cal_data_cfg=none
        ps_mode=1
        auto_ds=1
        host_mlme=1
        fw_name=nxp/sduart_nw61x_v1.bin.se
}
```

4. Load the drivers and firmware.

```
modprobe moal mod_para=nxp/wifi_mod_para.conf
```

WiFi

Bring up Wi-Fi interfaces

5. Verify that the firmware was downloaded successfully via driver logs on the console.

dmesg -w

```
[ 59.848148] mlan: loading out-of-tree module taints kernel.
[ 59.870831] wlan: Loading MWLAN driver
[ 59.871185] wlan: Register to Bus Driver...
[ 59.871381] vendor=0x0471 device=0x0205 class=0 function=1
[ 59.871472] Attach moal handle ops, card interface type: 0x109
[ 59.871483] rps set to 0 from module param
[ 59.872310] SDIW612: init module param from usr cfg
[ 59.872376] card_type: SDIW612, config block: 0
[ 59.872387] cfg80211_wext=0xf
[ 59.872390] max_vir_bss=1
[ 59.872395] cal_data_cfg=none
[ 59.872398] ps_mode = 1
[ 59.872400] auto_ds = 1
[ 59.872407] host_mlme=enable
[ 59.872411] fw_name=nxp/sduart_nw61x_v1.bin.se
[ 59.872432] SDIO: max_segs=128 max_seg_size=65535
[ 59.872437] rx_work=1 cpu_num=2
[ 59.872442] Enable moal_recv_amsdu_packet
[ 59.872461] Attach mlan adapter operations.card type is 0x109.
[ 59.872745] wlan: Enable TX SG mode
[ 59.872754] wlan: Enable RX SG mode
```

```
[ 59.872745] wlan: Enable TX SG mode
[ 59.872754] wlan: Enable RX SG mode
[ 59.880985] Request firmware: nxp/sduart_nw61x_v1.bin.se
[ 60.160989] Wlan: FW download over, firmwarelen=1065364
downloaded 953164
[ 60.588889] WLAN FW is active
[ 60.588903] on_time is 60412073569
[ 60.615581] VDLL image: len=112200
[ 60.616635] fw_cap_info=0x487cff03, dev_cap_mask=0xffffffff
[ 60.616654] uuid: 544906e672fd5192ac42d748be0e5109
[ 60.616659] max_p2p_conn = 8, max_sta_conn = 16
[ 60.738506] Register NXP 802.11 Adapter mlan0
[ 60.738599] wlan: uap%d set max_mtu 2000
[ 60.744185] Register NXP 802.11 Adapter uap0
[ 60.838830] Register NXP 802.11 Adapter wfd0
[ 60.838874] wlan: version = SDIW612---18.99.3.p21.10-
MM6X18505.p4-GPL-(FP92)
[ 60.868735] wlan: Register to Bus Driver Done
[ 60.868751] wlan: Driver loaded successfully
```

WiFi

Bring up Wi-Fi interfaces

6. Once the driver modules loaded successfully, verify the Wi-Fi interface created by the driver using below given command.

ifconfig -a

```
mlan0: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        ether 20:ba:36:5c:e6:c9  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions
0

uap0: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        ether 22:ba:36:5c:e7:c9  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions
0

wfd0: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        ether 22:ba:36:5c:e6:c9  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions
0
```

WiFi

Create STA Mode and connect to Ext AP

This section describes configuration for the STA mode on the FRDM i.MX 93 board using the **/etc/wpa_supplicant.conf** configuration file.

The wpa_supplicant utility is configured using a text file that lists all expected networks and security policies, including pre-shared keys.

1. Open configuration file /etc/wpa_supplicant.conf which includes the network details of the Ext-AP (Here Ext-AP could be your home/office Wi-Fi router or Mobile hotspot) that you want to connect to.

Sample wpa_supplicant.conf with WPA2 security:

```
$ nano /etc/wpa_supplicant.conf
```

```
ctrl_interface=/var/run/wpa_supplicant
ctrl_interface_group=0
update_config=1
network={
ssid="NXP_Demo" #modify
key_mgmt=WPA-PSK
psk="123456789" #modify
}
```

WiFi

Create STA Mode and connect to Ext AP

2. Bring up the mlan0 interface and verify the STA mode interface using following command.

```
ifconfig mlan0 up
ifconfig mlan0
```

```
mlan0: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        ether 20:ba:36:5c:e6:c9  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions
0
```

WiFi

Create STA Mode and connect to Ext AP

3. Make sure to stop any running instance of the wpa_supplicant to avoid failures.

```
killall wpa_supplicant
```

```
root@imx93frdm-iwxxx-matter:~# killall wpa_supplicant
root@imx93frdm-iwxxx-matter:~# killall wpa_supplicant
wpa_supplicant: no process found
```

WiFi

Create STA Mode and connect to Ext AP

4. Use following command to start wpa_supplicant in background.

$ wpa_supplicant -imlan0 -Dnl80211 -c /etc/wpa_supplicant.conf –B

```
root@imx93frdm-iwxxx-matter:~# wpa_supplicant -imlan0 -Dnl80211 -
c /etc/wpa_supplicant.conf -B
Successfully initialized wpa_supplicant
rfkill: Cannot open RFKILL control device
root@imx93frdm-iwxxx-matter:~# [ 1278.242273] wlan: mlan0 START
SCAN
[ 1283.505491] wlan: SCAN COMPLETED: scanned AP count=51
[ 1283.519298] wlan: HostMlme mlan0 send auth to bssid
be:XX:XX:XX:b0:40
[ 1283.522723] mlan0:
[ 1283.522736] wlan: HostMlme Auth received from
be:XX:XX:XX:b0:40 [ 1283.567080] wlan: HostMlme mlan0 Connected
to bssid be:XX:XX:XX:b0:40 successfully
[ 1283.582054] mlan0:
[ 1283.582069] wlan: Send EAPOL pkt to be:XX:XX:XX:b0:40
[ 1283.589059] mlan0:
[ 1283.589072] wlan: Send EAPOL pkt to be:XX:XX:XX:b0:40
[ 1283.596362] woal_cfg80211_set_rekey_data return:
gtk_rekey_offload is DISABLE
```

WiFi

Create STA Mode and connect to Ext AP

5. Once the connection is done successfully, start the udhcp client by using the following command
to get the dynamic IP address from the Ext-AP and it could be any IP address of the format “xxx.xxx.xxx.xxx”.

For example, in below snapshot the IP address “192.168.105.27” is obtained from the Ext-AP.

NOTE: Please wait for some time after executing the following command. On successful execution
of the command it will return to root prompt by pressing “enter” key. If it does not return to root
prompt, there is some failure.

```
$ udhcpc -i mlan0
```

```
root@imx93frdm-iwxxx-matter:~# udhcpc -i mlan0
udhcpc: started, v1.36.1
Dropped protocol specifier '.udhcpc' from 'mlan0.udhcpc'. Using
'mlan0' (ifindex=4).
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.105.27, server
192.168.105.68
udhcpc: lease of 192.168.105.27 obtained from 192.168.105.68,
lease time 3599
/etc/udhcpc.d/50default: Adding DNS 192.168.105.68
Dropped protocol specifier '.udhcpc' from 'mlan0.udhcpc'. Using
'mlan0' (ifindex=4).
root@imx93frdm-iwxxx-matter:~#
```

WiFi

Create STA Mode and connect to Ext AP

6. Verify IP address assigned to STA device interfac
For example, in below snapshot the IP address “192

$ ifconfig mlan0
$ ping 8.8.8.8
$ ping www.google.com

```
root@imx93frdm-iwxxx-matter:~# ifconfig mlan0
mlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.105.27  netmask 255.255.255.0  broadcast
192.168.105.255
        inet6 fe80::22ba:36ff:fe5c:e6c9  prefixlen 64  scopeid
0x20<link>
        ether 20:ba:36:5c:e6:c9  txqueuelen 1000  (Ethernet)
        RX packets 7  bytes 1212 (1.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 36  bytes 5328 (5.2 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions
0

root@imx93frdm-iwxxx-matter:~#
```

If fail log :
ping: google.com: Temporary failure in name resolution
Fix:

sudo sh -c 'cat >/etc/resolv.conf <<EOF
nameserver 8.8.8.8
nameserver 1.1.1.1
options timeout:2 attempts:2
EOF’
“

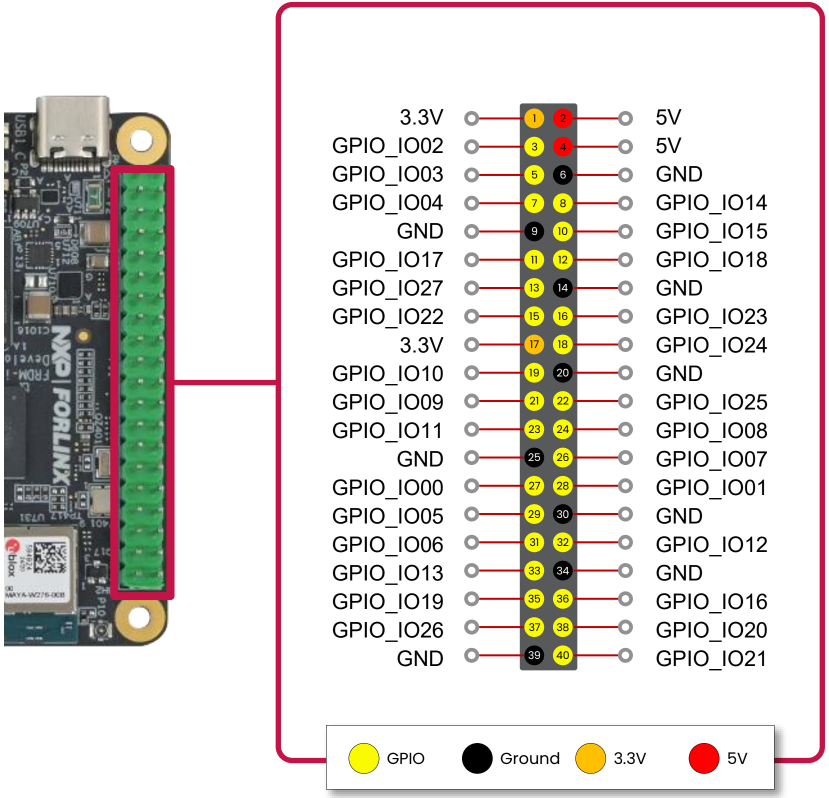
Basic IO control

Config GPIO

```
root@imx93frdm:/# cat /sys/kernel/debug/pinctrl/*/pins
registered pins: 108
pin 0 (IMX93_IOMUXC_DAP_TDI) 28:43820000.gpio 443c0000.pinctrl
pin 1 (IMX93_IOMUXC_DAP_TMS_SWDIO) 29:43820000.gpio 443c0000.pinctrl
pin 2 (IMX93_IOMUXC_DAP_TCLK_SWCLK) 30:43820000.gpio 443c0000.pinctrl
pin 3 (IMX93_IOMUXC_DAP_TDO_TRACESWO) 31:43820000.gpio 443c0000.pinctrl
pin 4 (IMX93_IOMUXC_GPIO_IO00) 0:43810000.gpio 443c0000.pinctrl
pin 5 (IMX93_IOMUXC_GPIO_IO01) 1:43810000.gpio 443c0000.pinctrl
pin 6 (IMX93_IOMUXC_GPIO_IO02) 2:43810000.gpio 443c0000.pinctrl
pin 7 (IMX93_IOMUXC_GPIO_IO03) 3:43810000.gpio 443c0000.pinctrl
```

```
root@imx93frdm:/# gpiodetect
gpiochip0 [43810000.gpio] (32 lines)
gpiochip1 [43820000.gpio] (32 lines)
gpiochip2 [43830000.gpio] (32 lines)
gpiochip3 [47400000.gpio] (32 lines)
gpiochip4 [1-0022] (24 lines)
gpiochip5 [0-0020] (8 lines)
```

```
root@imx93frdm:/# gpioset -i -c gpiochip0 2=1
gpioset> set 2=1
gpioset> get 2
"2"=active
gpioset> set 2=0
gpioset> get 2
"2"=inactive
gpioset>
gpioset> exit
root@imx93frdm:/#
```



Useful Link with FRDM i.MX93

NXP offer

- Getting Started with FRDM-IMX93 https://www.nxp.com/document/guide/getting-started-with-frdm-imx93:GS-FRDM-IMX93
- i.MX FRDM Software User Guide https://docs.nxp.com/bundle/UG10195/page/topics/introduction.html
- **Debian** documents at http://www.nxp.com/nxpdebian
- GoPoint repo: https://github.com/nxp-imx-support/nxp-demo-experience-demos-list
- Image Classification/Object Detection: https://github.com/nxp-imx/eiq-example
- i.MX Smart Fitness: https://github.com/nxp-imx-support/imx-smart-fitness
- i.MX Smart Kitchen: https://github.com/nxp-imx-support/smart-kitchen
- i.MX E-Bike VIT: https://github.com/nxp-imx-support/imx-ebike-vit
- eIQ toolkit : https://www.nxp.com/design/design-center/software/eiq-ml-development-environment/eiq-toolkit-for-end-to-end-model-development-and-deployment:EIQ-TOOLKIT

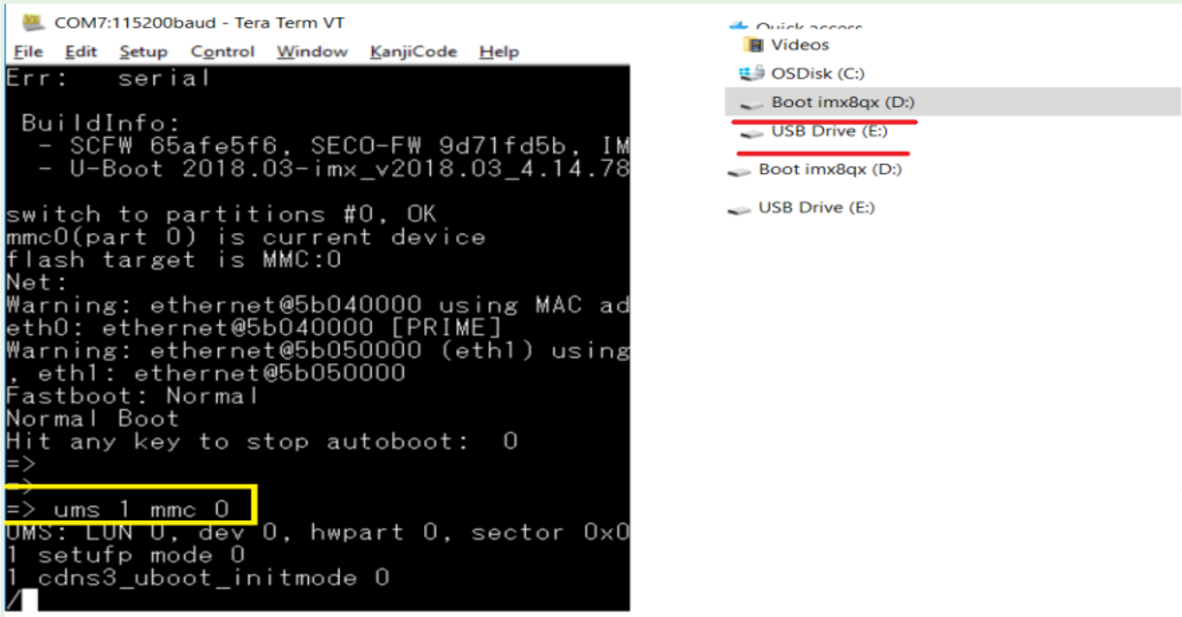
Thank you

Backup

UMS

How to upgrade Linux Kernel and dtb on eMMC without UUU

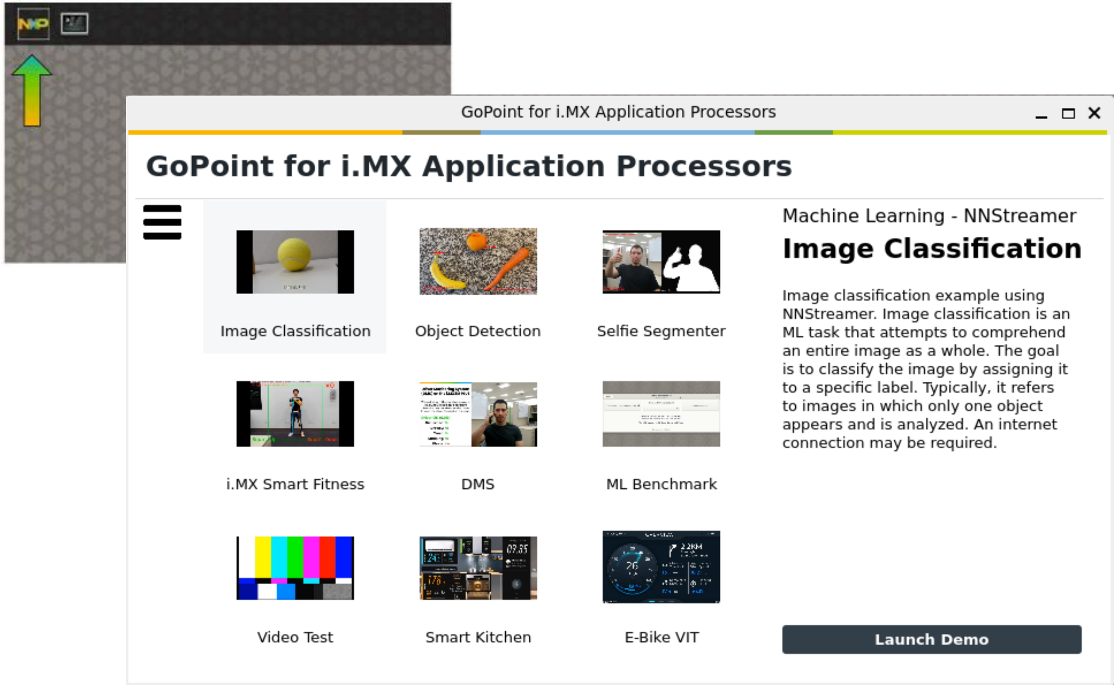
Usage:
ums <USB_controller> [<devtype>] <dev[:part]>
e.g. ums 1 mmc 0
devtype defaults to mmc



GoPoint

GoPoint Demo with FRDM i.MX93

Click on the NXP icon in the top left corner of the desktop to access the GoPoint Demo Menu

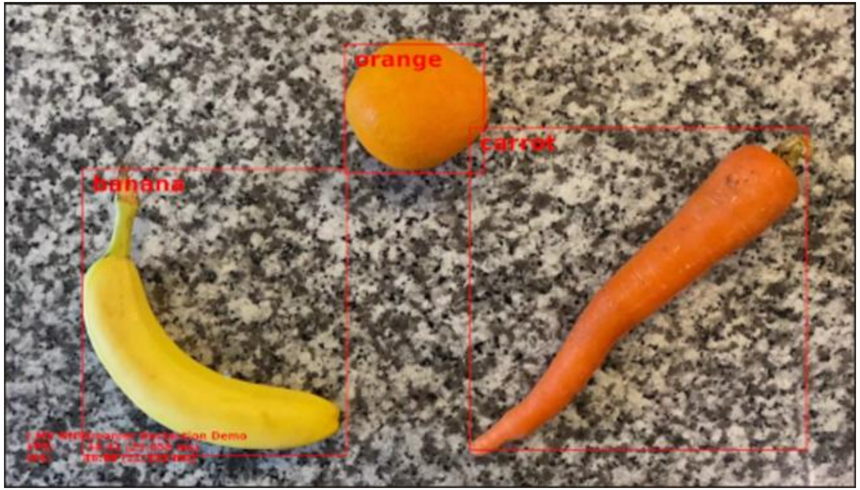


1. Image Classification
2. Object Detection
3. Selfie Segmenter
4. i.MX Smart Fitness
5. DMS (Driver Monitor System)
6. ML Benchmark
7. Video Test
8. i.MX Smart Kitchen
9. i.MX E-Bike VIT

Please connect to the internet.

GoPoint Demo with FRDM i.MX93

Object Detection

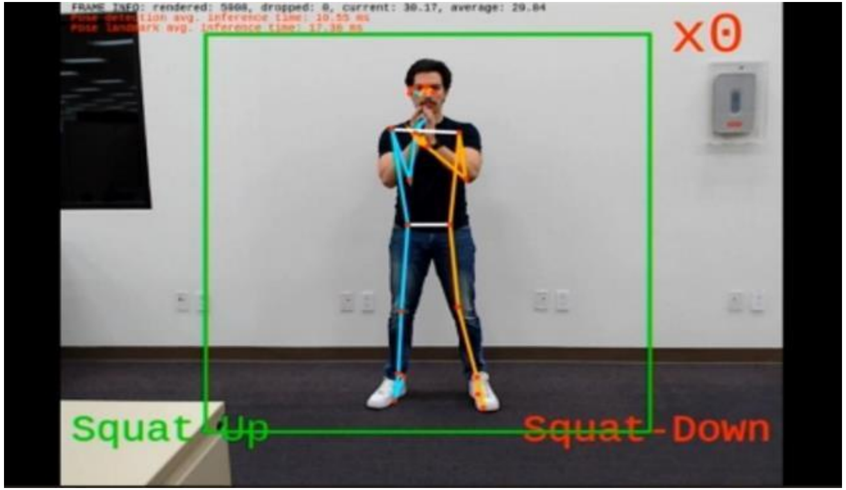


Selfie Segmenter

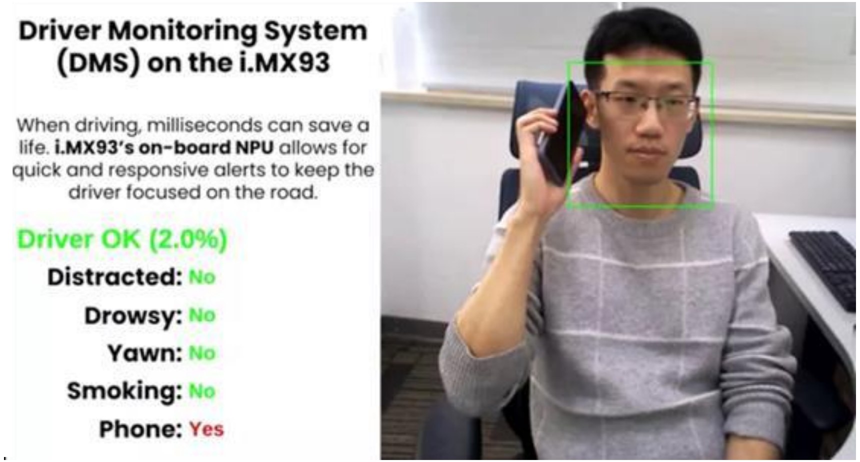
GoPoint Demo with FRDM i.MX93

AVNET

Smart Fitness



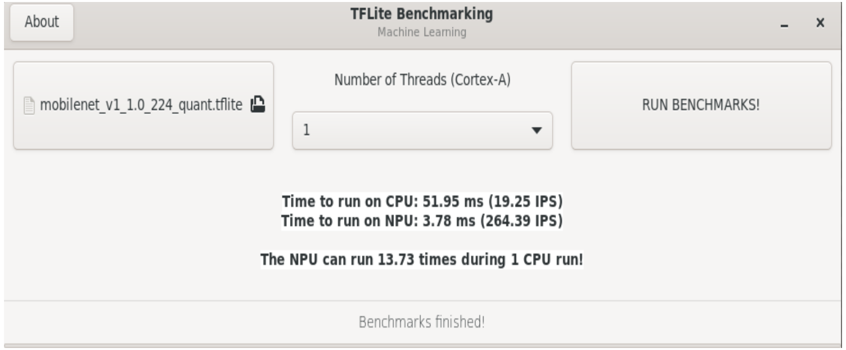
DMS



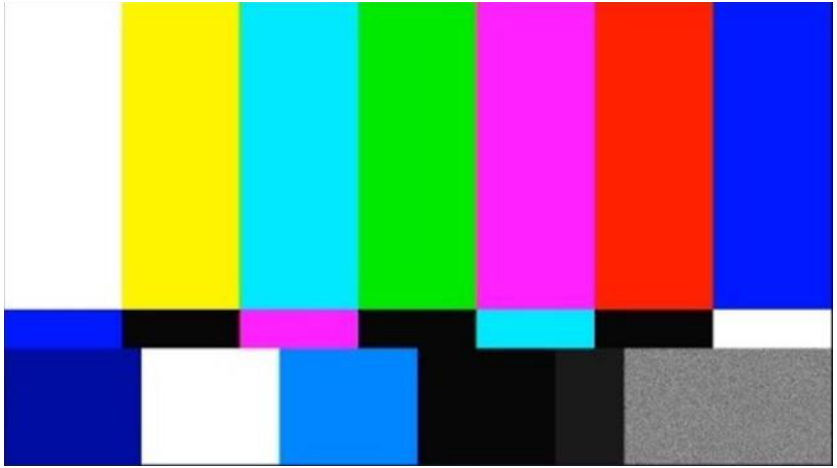
GoPoint Demo with FRDM i.MX93

AVNET

ML Benchmark



Video Test Demo



GoPoint Demo with FRDM i.MX93

Smart Kitchen

E-Bike VIT Demo